SDG 15.4.1

SDG 15 Life on Land

Water discharge guidelines and standards

Mariano Marcos State University · THE Sustainability Impact Ratings 2027

THE QUESTION

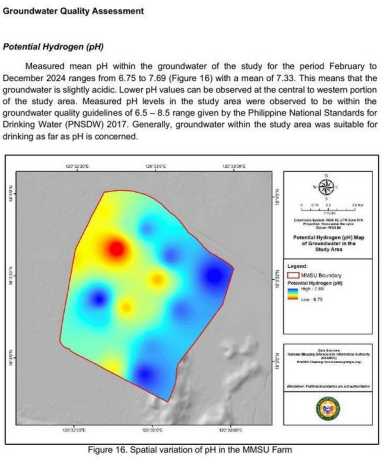
Have water quality standards and guidelines for water discharges (to uphold water quality in order to protect ecosystems, wildlife, and human health and welfare).

1 Introduction

In MMSU, the Water Policy, part of Sirmata 2040 (2021) and based on the Water Code of the Philippines, requires that all water and related resources of the university, ground and surface, be shielded from any kind of pollution (Policy 2.9) and that polluted resources be rehabilitated immediately, with attention to impoundments, catch basins and water treatment facilities (Policy 2.8). The Integrated Pollution and Waste Management Program, revised in 2020, provides for registration as a hazardous waste generator for wastewater from agro-industrial facilities such as the piggery, in compliance with the Clean Water Act (RA 9275). Memorandum No. 26-370 of 30 September 2026 adds a compliance programme covering wastewater, with DENR-EMB permits, monitoring and reporting. From February 2024 to June 2025, 17 wells on the MMSU Farm were also checked monthly for pH, dissolved solids, conductivity, chloride and nitrate-nitrogen.

2 Body

2.1 Groundwater Quality Monitoring at the MMSU Farm


Groundwater Quality Assessment

Potential Hydrogen (pH)

Measured mean pH within the groundwater of the study for the period February to December 2024 ranges from 6.75 to 7.69 (Figure 16) with a mean of 7.33. This means that the groundwater is slightly acidic. Lower pH values can be observed at the central to western portion of the study area. Measured pH levels in the study area were observed to be within the groundwater quality guidelines of 6.5 – 8.5 range given by the Philippine National Standards for Drinking Water (PNSDW) 2017. Generally, groundwater within the study area was suitable for drinking as far as pH is concerned.

Figure 16. Spatial variation of pH in the MMSU Farm

Document: Groundwater Quality Assessment of the MMSU Farm, spatial variation of pH (monitoring through June 2025).

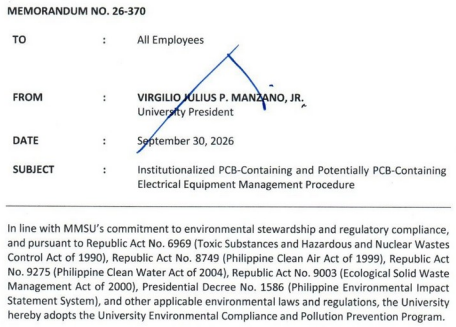
Through June 2025, MMSU continued monthly monitoring of 17 groundwater wells on the MMSU Farm, measuring pH, total dissolved solids, electrical conductivity, chloride and nitrate-nitrogen and comparing the results with the Philippine National Standards for Drinking Water (2017). The assessment, covering February 2024 to June 2025, flags wells with persistently high dissolved solids, which rose again from January to May 2025, for continued monitoring to guide water resource management and prevent further degradation of groundwater quality.

Mean pH from February to December 2024 ranged from 6.75 to 7.69, within the 6.5–8.5 guideline of the national drinking water standards. Small campus reservoirs also collect rainwater, reduce flooding and recharge groundwater.

RESOURCES 1

2.2 Memorandum No. 26-370 Institutionalizes MMSU's Environmental Compliance and Hazardous Waste Procedures

Ongoing programme · 30 September 2026



MEMORANDUM NO. 26-370

TO : All Employees

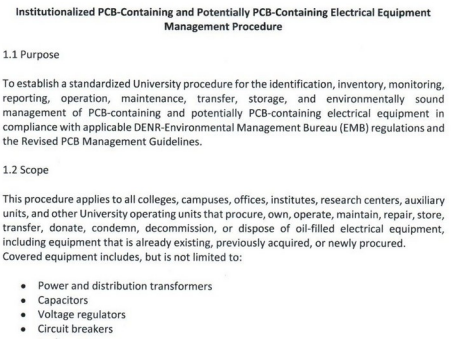
FROM : VIRGILIO JULIUS P. MANZANO, JR.
University President

DATE : September 30, 2026

SUBJECT : Institutionalized PCB-Containing and Potentially PCB-Containing Electrical Equipment Management Procedure

In line with MMSU's commitment to environmental stewardship and regulatory compliance, and pursuant to Republic Act No. 6969 (Toxic Substances and Hazardous and Nuclear Wastes Control Act of 1990), Republic Act No. 8749 (Philippine Clean Air Act of 1999), Republic Act No. 9275 (Philippine Clean Water Act of 2004), Republic Act No. 9003 (Ecological Solid Waste Management Act of 2000), Presidential Decree No. 1586 (Philippine Environmental Impact Statement System), and other applicable environmental laws and regulations, the University hereby adopts the University Environmental Compliance and Pollution Prevention Program.

Document: Office of the President Memorandum No. 26-370, 30 September 2026, adopting the University Environmental Compliance and Pollution Prevention Program.



Institutionalized PCB-Containing and Potentially PCB-Containing Electrical Equipment Management Procedure

1.1 Purpose

To establish a standardized University procedure for the identification, inventory, monitoring, reporting, operation, maintenance, transfer, storage, and environmentally sound management of PCB-containing and potentially PCB-containing electrical equipment in compliance with applicable DENR-Environmental Management Bureau (EMB) regulations and the Revised PCB Management Guidelines.

1.2 Scope

This procedure applies to all colleges, campuses, offices, institutes, research centers, auxiliary units, and other University operating units that procure, own, operate, maintain, repair, store, transfer, donate, condemn, decommission, or dispose of oil-filled electrical equipment, including equipment that is already existing, previously acquired, or newly procured.
Covered equipment includes, but is not limited to:

- Power and distribution transformers
- Capacitors
- Voltage regulators
- Circuit breakers

Document: attached procedure for PCB-containing and potentially PCB-containing electrical equipment (purpose, scope and inventory), Memorandum No. 26-370.

Policy	Memorandum No. 26-370, University Environmental Compliance and Pollution Prevention Program
Approved by	University President Virgilio Julius P. Manzano Jr.
Reference no.	26-370
Date	30 September 2026
Provisions	Environmental Compliance Procedures; PCB-containing equipment procedure

On September 30, 2026, University President Virgilio Julius P. Manzano Jr. issued Memorandum No. 26-370 to all employees, adopting the University Environmental Compliance and Pollution Prevention Program under Republic Act 6969 (Toxic Substances and Hazardous and Nuclear Wastes Control Act), Republic Act 8749 (Clean Air Act), Republic Act 9275 (Clean Water Act), Republic Act 9003 (Ecological Solid Waste Management Act) and Presidential Decree 1586 (Environmental Impact Statement System). The memorandum covers the management of hazardous wastes, wastewater, air emissions, solid wastes and chemicals, together with permits, monitoring, reporting and inspections required by the Department of Environment and Natural Resources – Environmental Management Bureau (DENR-EMB), and directs all offices, colleges, laboratories, research units and auxiliary enterprises to comply with the attached Environmental Compliance Procedures.

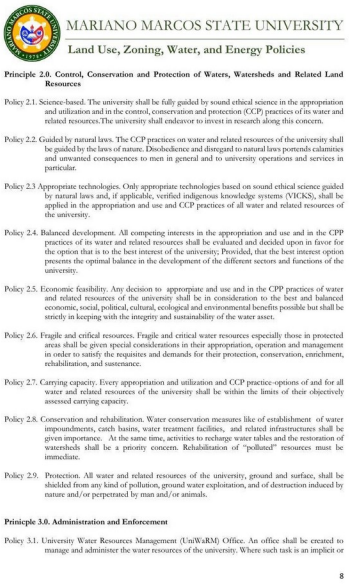
The attached procedure for PCB-containing and potentially PCB-containing electrical equipment, such as transformers, capacitors, voltage regulators, circuit breakers, reclosers and oil-filled switches, requires every unit to inventory its equipment and submit the inventory to the Sustainability and Futures Thinking (SFT) Office. The SFT Office validates the inventory, keeps the university's central PCB inventory and the DENR-EMB online PCB

inventory, ensures labelling, conducts inspections and annual inventory validation, and prepares the reports required by DENR-EMB. No covered equipment may be relocated, transferred, repaired through oil handling, decommissioned, condemned or disposed of without the SFT Office's written clearance, and PCB wastes and equipment may be disposed of only through DENR-accredited transporters, treaters and disposal facilities. The SFT Office keeps hazardous waste manifests and certificates of treatment or disposal among its records.

RESOURCES 2

2.3 MMSU Water Policy Protects University Waters from Pollution

Policy · Sirmata 2040 (2021) and Integrated Pollution and Waste Management Program (revised 2020)



MARIANO MARCOS STATE UNIVERSITY

Land Use, Zoning, Water, and Energy Policies

Principle 2.0. Control, Conservation and Protection of Waters, Watersheds and Related Land Resources

Policy 2.1. Science-based. The university shall be fully guided by sound ethical science in the appropriation and utilization and in the control, conservation and protection (CCP) practices of its water and related resources.The university shall endeavor to invest in research along this concern.

Policy 2.2. Guided by natural laws. The CCP practices on water and related resources of the university shall be guided by the laws of nature. Disobedience and disregard to natural laws portends calamities and unwanted consequences to men in general and to university operations and services in particular.

Policy 2.3 Appropriate technologies. Only appropriate technologies based on sound ethical science guided by natural laws and, if applicable, verified indigenous knowledge systems (VICKS), shall be applied in the appropriation and use and CCP practices of all water and related resources of the university.

Policy 2.4. Balanced development. All competing interests in the appropriation and use and in the CPP practices of its water and related resources shall be evaluated and decided upon in favor for the option that is to the best interest of the university; Provided, that the best interest option presents the optimal balance in the development of the different sectors and functions of the university.

Policy 2.5. Economic feasibility. Any decision to appropriate and use and in the CPP practices of water and related resources of the university shall be in consideration to the best and balanced economic, social, political, cultural, ecological and environmental benefits possible but shall be strictly in keeping with the integrity and sustainability of the water asset.

Policy 2.6. Fragile and critical resources. Fragile and critical water resources especially those in protected areas shall be given special considerations in their appropriation, operation and management in order to satisfy the requisites and demands for their protection, conservation, enrichment, rehabilitation, and sustenance.

Policy 2.7. Carrying capacity. Every appropriation and utilization and CCP practice-options of and for all water and related resources of the university shall be within the limits of their objectively assessed carrying capacity.

Policy 2.8. Conservation and rehabilitation. Water conservation measures like of establishment of water impoundments, catch basins, water treatment facilities, and related infrastructures shall be given importance. At the same time, activities to recharge water tables and the restoration of watersheds shall be a priority concern. Rehabilitation of "polluted" resources must be immediate.

Policy 2.9. Protection. All water and related resources of the university, ground and surface, shall be shielded from any kind of pollution, ground water exploitation, and of destruction induced by nature and/or perpetrated by man and/or animals.

Principle 3.0. Administration and Enforcement

Policy 3.1. University Water Resources Management (UniWaRM) Office. An office shall be created to manage and administer the water resources of the university. Where such task is an implicit or

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Document: MMSU Water Policy, control, conservation and protection of waters (Policies 2.6–2.9), Sirmata 2040 LUDIP.

Policy	MMSU Water Policy (Sirmata 2040 LUDIP) and Integrated Pollution and Waste Management Program
Date	Water Policy 2021; IPWMP revised 2020.
Provisions	Water Policy 2.6–2.9 (2.9 protection from pollution; 2.8 rehabilitation); IPWMP registration for wastewater discharge (RA 9275)

MMSU's Water Policy under Sirmata 2040, based on the Water Code of the Philippines and its 2005 implementing rules, requires that "all water and related resources of the university, ground and surface, shall be shielded from any kind of pollution" (Policy 2.9) and that the rehabilitation of polluted resources be immediate, giving importance to water impoundments, catch basins and water treatment facilities (Policy 2.8).

The university's Integrated Pollution and Waste Management Program also provides for registration as a hazardous waste generator for wastewater discharge from agro-industrial facilities such as the piggery, in compliance with Republic Act 9275, the Philippine Clean Water Act.

RESOURCES 3, 4, 5

3 Conclusion

The Water Policy shields university waters from pollution, the pollution programme provides for registering wastewater from agro-industrial facilities, and 17 farm wells are monitored month by month.

4 Resources

1. MMSU, Groundwater Quality Assessment of the MMSU Farm (monitoring February 2024 – June 2025). https://www.mmsu.edu.ph/resources/15g9Xtys7q3Rujd0-CZvPH4OHTdJ1__Z8/data
2. MMSU Office of the President, Memorandum No. 26-370, “Institutionalized PCB-Containing and Potentially PCB-Containing Electrical Equipment Management Procedure,” 30 September 2026
3. MMSU, Land Use, Zoning, Water and Energy Policies (Sirmata 2040 LUDIP, 2021). https://www.mmsu.edu.ph/resources/19uycXrgcO_iXPdPpMQwdCRxmoD1U5i60/view
4. MMSU, Revised Integrated Pollution and Waste Management Program. https://www.mmsu.edu.ph/resources/1fNUCZwyOozOKO27L2Bm6pNN-P_cJNpLd/view
5. MMSU, Water Discharge Management (SDG 15 page). https://www.mmsu.edu.ph/resources/1l8foDfenNc7xpKp8wM8XShfZT58YyjSH/data